```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
df=pd.read_csv('mail_data.csv')
df
```

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...
...	...	...
5567	spam	This is the 2nd time we have tried 2 contact u...
5568	ham	Will ü b going to esplanade fr home?
5569	ham	Pity, * was in mood for that. So...any other s...
5570	ham	The guy did some bitching but I acted like i'd...
5571	ham	Rofl. Its true to its name

5572 rows × 2 columns

Next steps: Generate code with df New interactive sheet

```
data=df.where((pd.notnull(df)),'')
data
```

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...
...	...	...
5567	spam	This is the 2nd time we have tried 2 contact u...
5568	ham	Will ü b going to esplanade fr home?
5569	ham	Pity, * was in mood for that. So...any other s...
5570	ham	The guy did some bitching but I acted like i'd...
5571	ham	Rofl. Its true to its name

5572 rows × 2 columns

Next steps: Generate code with data New interactive sheet

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5572 entries, 0 to 5571
Data columns (total 2 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Category  5572 non-null   object
 1   Message   5572 non-null   object
dtypes: object(2)
memory usage: 87.2+ KB
```

```
data.shape
```

```
(5572, 2)
```

```
from pandas.core.indexes import category
data.loc[data['Category']=='spam','Category']=0
data.loc[data['Category']=='ham','Category']=1
```

```
X=data['Message']
Y=data['Category']
```

```
X
```

	Message
0	Go until jurong point, crazy.. Available only ...
1	Ok lar... Joking wif u oni...
2	Free entry in 2 a wkly comp to win FA Cup fina...
3	U dun say so early hor... U c already then say...
4	Nah I don't think he goes to usf, he lives aro...
...	...
5567	This is the 2nd time we have tried 2 contact u...
5568	Will ü b going to esplanade fr home?
5569	Pity, * was in mood for that. So...any other s...
5570	The guy did some bitching but I acted like i'd...
5571	Rofl. Its true to its name

5572 rows × 1 columns

dtype: object

```
Y
```

	Category
0	1
1	1
2	0
3	1
4	1
...	...
5567	0
5568	1
5569	1
5570	1
5571	1

5572 rows × 1 columns

dtype: object

```
X_train,X_test,Y_train,Y_test=train_test_split(X,Y,test_size=0.2,random_state=3)
```

```
print(X.shape)
print(X_train.shape)
print(X_test.shape)
print(Y.shape)
print(Y_train.shape)
print(Y_test.shape)
```

```
(5572,)
(4457,)
(1115,)
(5572,)
(4457,)
(1115,)
```

```
feature_extraction = TfidfVectorizer(min_df = 1, stop_words='english', lowercase=True)
X_train_features = feature_extraction.fit_transform(X_train)
X_test_features = feature_extraction.transform(X_test)
```

```
Y_train = Y_train.astype('int')
Y_test = Y_test.astype('int')
```

```
print(X_train)
```

```
3075                  Don know. I did't msg him recently.
1787    Do you know why god created gap between your f...
1614                          Thnx dude. u guys out 2nite?
4304                                        Yup i'm free...
3266    44 7732584351, Do you want a New Nokia 3510i c...
                              ...
789     5 Free Top Polyphonic Tones call 087018728737,...
968     What do u want when i come back?.a beautiful n...
1667    Guess who spent all last night phasing in and ...
3321    Eh sorry leh... I din c ur msg. Not sad alread...
1688    Free Top ringtone -sub to weekly ringtone-get ...
Name: Message, Length: 4457, dtype: object
```

```
print(X_train_features)
```

```
<Compressed Sparse Row sparse matrix of dtype 'float64'
        with 34775 stored elements and shape (4457, 7431)>
  Coords        Values
  (0, 2329)     0.38783870336935383
  (0, 3811)     0.34780165336891333
  (0, 2224)     0.413103377943378
  (0, 4456)     0.4168658090846482
  (0, 5413)     0.6198254967574347
  (1, 3811)     0.17419952275504033
  (1, 3046)     0.2503712792613518
  (1, 1991)     0.33036995955537024
  (1, 2956)     0.33036995955537024
  (1, 2758)     0.3226407885943799
  (1, 1839)     0.2784903590561455
  (1, 918)      0.22871581159877646
  (1, 2746)     0.3398297002864083
  (1, 2957)     0.3398297002864083
  (1, 3325)     0.31610586766078863
  (1, 3185)     0.29694482957694585
  (1, 4080)     0.18880584110891163
  (2, 6601)     0.6056811524587518
  (2, 2404)     0.45287711070606745
  (2, 3156)     0.4107239318312698
  (2, 407)      0.509272536051008
  (3, 7414)     0.8100020912469564
  (3, 2870)     0.5864269879324768
  (4, 2870)     0.41872147309323743
  (4, 487)      0.2899118421746198
  :     :
  (4454, 2855)  0.47210665083641806
  (4454, 2246)  0.47210665083641806
  (4455, 4456)  0.24920025316220423
  (4455, 3922)  0.31287563163368587
  (4455, 6916)  0.19636985317119715
  (4455, 4715)  0.30714144758811196
  (4455, 3872)  0.3108911491788658
  (4455, 7113)  0.30536590342067704
  (4455, 6091)  0.23103841516927642
  (4455, 6810)  0.29731757715898277
  (4455, 5646)  0.33545678464631296
  (4455, 2469)  0.35441545511837946
  (4455, 2247)  0.37052851863170466
  (4456, 2870)  0.31523196273113385
  (4456, 5778)  0.16243064490100795
  (4456, 334)   0.2220077711654938
  (4456, 6307)  0.2752760476857975
  (4456, 6249)  0.17573831794959716
  (4456, 7150)  0.3677554681447669
  (4456, 7154)  0.24083218452280053
  (4456, 6028)  0.21034888000987115
  (4456, 5569)  0.4619395404299172
  (4456, 6311)  0.30133182431707617
  (4456, 647)   0.30133182431707617
  (4456, 141)   0.292943737785358
```

```
model=LogisticRegression()
model.fit(X_train_features,Y_train)
```

```
▾ LogisticRegression
LogisticRegression()
```

```
prediction_train=model.predict(X_train_features)
accuracy_train=accuracy_score(Y_train,prediction_train)
```

```
print("ACC on training data:",accuracy_train)
```

```
ACC on training data: 0.9676912721561588
```

```
prediction_test=model.predict(X_test_features)
accuracy_test=accuracy_score(Y_test,prediction_test)
print("ACC on testing data:",accuracy_test)
```

```
ACC on testing data: 0.9668161434977578
```

```
input_mail=["""Subject: CONGRATULATIONS!!! You’ve WON $1,000,000 – Claim NOW!!!

Dear Lucky Winner,

We are thrilled to inform you that your email address has been randomly selected as the GRAND PRIZE WINNER of ONE MILLION DO

To claim your prize:
- Click this exclusive link: http://fake-prize-claim-now.com
- Enter your personal details (name, address, bank account, password)
- Act FAST – this offer expires in 24 HOURS!!!

Failure to respond immediately will result in forfeiting your winnings. Don’t miss this once-in-a-lifetime opportunity!!!

Sincerely,
International Lottery Promotions Team  """]

input_data_feature=feature_extraction.transform(input_mail)
prediction=model.predict(input_data_feature)
```

```
print(prediction)
if(prediction[0]==1):
    print("HAM MAIL")
else:
    print("SPAM MAIL")
```

```
[0]
SPAM MAIL
```

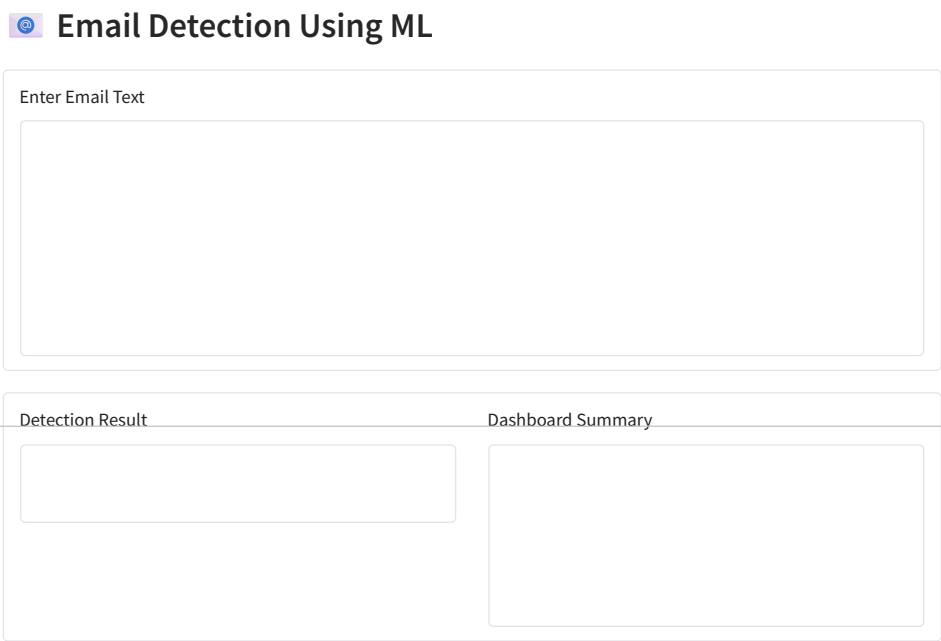
```
!pip install gradio
```

```
Requirement already satisfied: gradio in /usr/local/lib/python3.12/dist-packages (5.50.0)
Requirement already satisfied: aiofiles<25.0,>=22.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (24.1.0)
Requirement already satisfied: anyio<5.0,>=3.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (4.12.0)
Requirement already satisfied: brotli>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (1.2.0)
Requirement already satisfied: fastapi<1.0,>=0.115.2 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.123.10)
Requirement already satisfied: ffmpy in /usr/local/lib/python3.12/dist-packages (from gradio) (1.0.0)
Requirement already satisfied: gradio-client==1.14.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (1.14.0)
Requirement already satisfied: groovy~=0.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.1.2)
Requirement already satisfied: httpx<1.0,>=0.24.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.28.1)
Requirement already satisfied: huggingface-hub<2.0,>=0.33.5 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.36.0
Requirement already satisfied: jinja2<4.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (3.1.6)
Requirement already satisfied: markupsafe<4.0,>=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (3.0.3)
Requirement already satisfied: numpy<3.0,>=1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.0.2)
Requirement already satisfied: orjson~=3.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (3.11.5)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from gradio) (25.0)
Requirement already satisfied: pandas<3.0,>=1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.2.2)
Requirement already satisfied: pillow<12.0,>=8.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (11.3.0)
Requirement already satisfied: pydantic<=2.12.3,>=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.12.3)
Requirement already satisfied: pydub in /usr/local/lib/python3.12/dist-packages (from gradio) (0.25.1)
Requirement already satisfied: python-multipart>=0.0.18 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.0.20)
Requirement already satisfied: pyyaml<7.0,>=5.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (6.0.3)
Requirement already satisfied: ruff>=0.9.3 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.14.9)
Requirement already satisfied: safehttpx<0.2.0,>=0.1.6 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.1.7)
Requirement already satisfied: semantic-version~=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.10.0)
Requirement already satisfied: starlette<1.0,>=0.40.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.50.0)
Requirement already satisfied: tomlkit<0.14.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.13.3)
Requirement already satisfied: typer<1.0,>=0.12 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.20.0)
Requirement already satisfied: typing-extensions~=4.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (4.15.0)
Requirement already satisfied: uvicorn>=0.14.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.38.0)
Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from gradio-client==1.14.0->gradio) (2025.
Requirement already satisfied: websockets<16.0,>=13.0 in /usr/local/lib/python3.12/dist-packages (from gradio-client==1.14.0
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5.0,>=3.0->gradio) (3.11)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.115.2->
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio) (2025.11
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio) (1
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1.0,>=0.24.1-
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->gradi
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->gradi
Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->g
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->gra
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->gradio) (2025
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->gradio) (20
```

```
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=2.
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=2.0
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=
Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (8.3.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (13.
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas<3.0,
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=10.11.0->typer<1
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=10.11.0->typer
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->huggingfa
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->huggingface-hub
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=10.1
```

```
import gradio as gr

# Dashboard counters
total_scanned = 0
spam_count = 0
ham_count = 0

def classify_email(email_text):
    global total_scanned, spam_count, ham_count

    email_features = feature_extraction.transform([email_text])
    prediction = model.predict(email_features)

    total_scanned += 1

    if prediction[0] == 1:
        ham_count += 1
        result = "✅ HAM MAIL (Safe)"
    else:
        spam_count += 1
        result = "🚫 SPAM MAIL"

    summary = (
    f"📊 DASHBOARD SUMMARY\n\n"
    f"📧 Total Emails Scanned : {total_scanned}\n\n"
    f"🚫 Spam Emails         : {spam_count}\n\n"
    f"✅ Safe Emails         : {ham_count}"
)


    return result, summary


with gr.Blocks() as demo:
    gr.Markdown("# 🔍 Email Detection Using ML")

    email_input = gr.Textbox(lines=8, label="Enter Email Text")
    with gr.Row():
        result_output = gr.Textbox(
            label="Detection Result",
            lines=2
        )
        summary_output = gr.Textbox(
            label="Dashboard Summary",
            lines=6,
            interactive=False
        )

    scan_button = gr.Button("Scan Email")
    scan_button.click(
        classify_email,
        inputs=email_input,
        outputs=[result_output, summary_output]
    )

demo.launch()
```

```
It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting `share

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://aed8218bd0caf38136.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the
```



Next steps: Deploy to Cloud Run